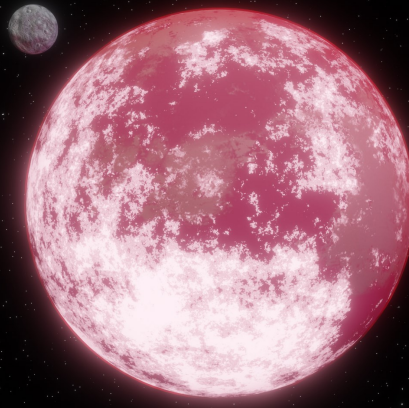


SAVANNAH CHAPPUS

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📍 Flagstaff, Arizona



WORK EXPERIENCE

JANUARY 2025 – MAY 2025

Computer Science Capstone Mentor | Northern Arizona University | Flagstaff, Arizona

- Mentored 3 small teams of undergraduate students, advising and supporting them as they designed and built various projects for their respective clients.
- Facilitated communication between capstone groups, clients, and the advising faculty.
- Coordinated weekly meetings and check-ins with each group, ensuring that they managed time and assignments efficiently and that conducive group dynamics were maintained.

JANUARY 2025 – MAY 2025

Graduate Teaching Assistant | Northern Arizona University | Flagstaff, Arizona

- Collaborated with another graduate teaching assistant to manage a group of 6 undergraduate graders, including schedules, deadlines, quality of work, and expectations.
- Crafted supplementary materials to support the efforts of the undergraduate graders.
- Resolved conflicts and issues when necessary.
- Supported both the professor and the graders by demonstrating grading expectations and ensuring assignments were graded in a timely manner.

JANUARY 2023 – MAY 2023, APRIL 2024 – present

Student Researcher | Northern Arizona University Department of Astronomy and Planetary Sciences | Flagstaff, Arizona

- Collaborated with a small group of researchers to develop programs for the Solar System Notification Alert Processing System (SNAPS).
- Assisted in instructing 15 students attending a week-long astroinformatics bootcamp hosted by the SNAPS team to conduct sprint-style research and recruit new researchers to the SNAPS team.
- Analyzed data from over 32,000 asteroids observed by the Zwicky Transient Facility (ZTF) telescope to formulate algorithms for detecting changes in behavior of individual asteroids.
- Designed and developed the Anomaly Detection for Asteroid Patterns and Trends (ADAPT) processing pipeline, designed to detect anomalies in individual asteroids using various machine learning algorithms.
- Wrote a research paper elaborating details of the design and development of the ADAPT pipeline.

OCTOBER 2023 – MAY 2024

Ecological Programmer | United States Geological Survey (employed through Northern Arizona University) | Remote

- Contributed research and programmatic development to SOILWAT2: an ecosystem water balance simulation model.
- Programmed solutions to significant new concepts like the spin-up capability in C for the simulation, which included updating the SOILWAT2 submodule for R version of the simulation and implementing additional changes to the R simulation as needed.
- Managed file changes and source control both locally and remotely using git and GitHub and adhered to the project's git standards and testing.
- Detected and debugged issues in both newly written code as well as in older code as the new code required.

AUGUST 2021 – MAY 2023

Peer Tutor | Northern Arizona University Academic Success Centers | Flagstaff, Arizona

- Provided group and individual assistance to over 500 unique students in high level mathematics content with over 2000 total visits from students.
- Collaborated with peers to facilitate group test reviews.
- Assisted students in developing good study skills and habits.
- Maintained attendance records of daily attendees.
- Developed resources and materials to ensure student success in math.



@niteflyunicorns.animation



Savannah Chappus



niteflyunicorns



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BIO

Hi, my name is Savannah, and I am an animator and designer who focuses on 3D animation, visualization, simulation, UI/UX design, and logo design. My background in mathematics and computer science give me a unique perspective on the animation process, and I enjoy the technical side of animation. I tend to lean towards more modern sleek designs, and I enjoy animating fun creatures and everything space related.

EDUCATION

Master of Science: Computer Science

May 2025 — Northern Arizona University

Bachelor of Fine Arts: Visual Communication (Motion Design emphasis)

May 2023 — Northern Arizona University

Bachelor of Science in Mathematics

May 2023 — Northern Arizona University



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PROJECTS

Mealie | November 2024

An open source, self-hosted recipe management platform (Vue.js, Python), <https://github.com/mealie-recipes/mealie>

KHES & SGAFT Logos | May 2023

Updated Peter Konjoian's existing logo to meet new business decisions and created a new logo for an upcoming project

Foodicons | September 2021

Designed several food/agriculture based icons which were selected as part of a crowd sourced icon design project organized by The Noun Library and The Lexicon

TXG Logo | February 2021

Logo redesign and update for the Tribal Exchange Network Group (TXG)

RELEVANT COURSES

- Augmented Reality
- GPU Programming
- Computer Graphics
- Virtual Worlds

ACTIVITIES & VOLUNTEERING

High Altitude Roller Derby

September 2024 — present

Active league member, participating in 2-hour practices twice a week, serving on several committees, including sponsorship, design, and events, and volunteering, including helping with practices, setting up and running games, and public outreach.

American Institute of Graphic Arts

August 2019 — May 2022

Member: occasional attendance to meetings to participate in portfolio reviews and discussions.

REFERENCES

Dana Ernst

Professor | dana.ernst@nau.edu

Mike Gowanlock

Advisor | michael.gowanlock@nau.edu

David Trilling

Advisor | david.trilling@nau.edu

SKILLS

ANIMATION & DESIGN

	Blender design, lighting, rigging, animation
	Cinema 4D 3D modeling, environment design, animation
	Unreal Engine environment design
	Adobe After Effects compositing, 2D animation
	Toon Boom Harmony 2D animation
	Adobe InDesign print & document design
	Procreate digital drawing
	Adobe Illustrator general 2D design
	Adobe Photoshop general photo compositing
	Adobe XD UI/UX design and prototyping
	Unity game and software design and development
	Figma UI/UX design and prototyping

COMPUTING

	Python machine learning, data science
	C / C++ computer graphics, general computing
	EMACS text and code editing
	GIT file organization, tracking, storage, and sharing
	Java app design, general computing
	XML UI design
	SQL database design
	MIPS machine-level computing
	VIM code editing
	Android Studio app development, visual scripting
	HTML / CSS personal website development & design
	Vue.js website design

OTHER EXPERIENCE

Graduate Research in Astroinformatics

Northern Arizona University | January 2025 — May 2025

- Designed and developed the Anomaly Detection for Asteroid Patterns and Trends (ADAPT) processing pipeline for detecting anomalous behavior in individual asteroids. This detection pipeline utilizes machine learning algorithms like DBSCAN and Isolation Forest to filter out observations with anomalous readings across various features.
- Documented the research process, findings, and remaining questions in a formal research paper.

Graduate Research in Combinatorial Game Theory

Northern Arizona University | January 2025 — May 2025

- Analyzed a 3-dimensional 4-in-a-row game called Yamma, first released in summer of 2024. This included analyzing the physical game board, computing values for larger and smaller boards, writing code to play the game digitally, and creating a program designed to compute and analyze the gamegraph for the game.
- Documented the research process, findings, and remaining questions in a formal research paper.

